

# WASP-177b

R-2398

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-177\_200820 · created 2026-08-21T00:16:47+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459081.7514 +/- 0.0056 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.169 +/- 0.049                 |
| Transit depth (Rp/Rs)^2                           | 2.86 +/- 1.66 [%]               |
| Orbital Inclination (inc)                         | 87.43 +/- 2.89                  |
| Airmass coefficient 1 (a1)                        | 2.04 +/- 1.43                   |
| Airmass coefficient 2 (a2)                        | -0.11 +/- 0.42                  |
| Scatter in the residuals of the lightcurve fit is | 69.46 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 1.11                            |
| Optimal Annulus                                   | 6.5                             |
| Transit Duration (day)                            | 0.101 +/- 0.032                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.169 ± 0.049 vs 0.1618 (deviation 0.12σ) |
| Duration fit vs published | 0.101 d vs 0.0721 d (deviation 0.74σ)     |
| Mid-time O–C              | -13.9 min                                 |
| Depth SNR                 | 2.8                                       |
| Residual scatter          | 69.46 %                                   |

### Light curve

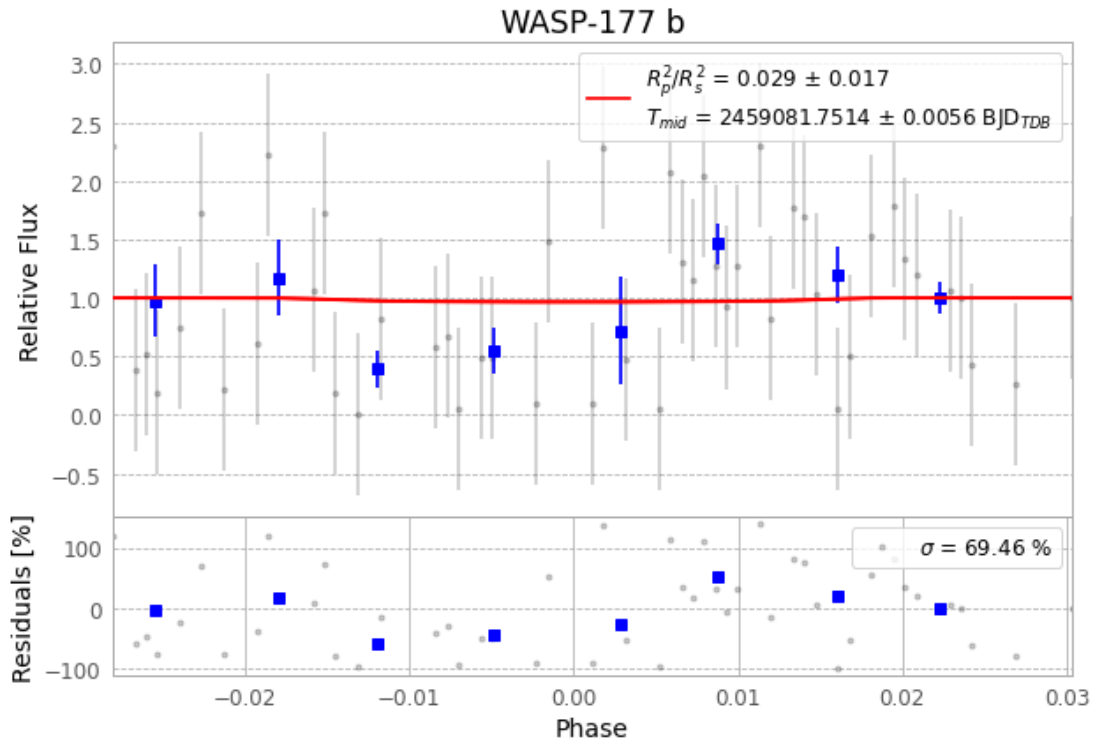


Figure 1 — FinalLightCurve\_WASP-177 b\_2020-08-19.png (SHA-256 b60ea94f524379d4...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [241, 329], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 8b4b1b29d2cd2322...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

87 FITS frames (57 MB), WASP-177200820034812.FITS → WASP-177200820080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-200820.log (9753f70e13ac...), FinalLightCurve\_WASP-177 b\_2020-08-19.png (b60ea94f5243...), FinalLightCurve\_WASP-177 b\_2020-08-19.csv (5cb254b6533d...)

## Compute resources

Wall time 164.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:16Z.